Given these two relations:
CUSTOMER (<u>cust_id</u>, cust_name, cust_address)
ORDER (<u>order_id</u>, order_date, cust_id)
If a customer may have any number of orders and each order is placed by a single customer, define the following terms *based on the above relations*:

1. Relation
    - A relation is a table that organizes data into rows and columns with details on the connection/relationship between the data
    - CUSTOMER is a relation with columns cust_id, cust_name and cust_address
    - ORDER is a relation that has a relationship with customer with columns order_id, order_date and cust_id
2. Attribute
    - An attribute is a column that represents a characteristic or a property of an entity.
    - In the CUSTOMER relation, the attributes are cust_id, cust_name and cust_address.
3. Domain
    - Domain refers to the set of values an attribute can take
    - For the cust_name attribute, the domain may be a set of strings representing the customer name.
    - For the order_date attribute, the domain may be a set of valid dates
    - For the cust_id or order_id attribute, the domain may be a set of unique numeric or alphanumeric characters representing the IDs
4. Tuple
    - A tuple is a single row in the a relation. It represents the set of values for the attributes of the relation.
    - For example a tuple in the CUSTOMER relation may look like (“CU0001”, “Bob”, “10 Street Avenue”
5. Degree and cardinality of a relation
    - Degree
        - The degree of a relation is the number of attributes (column) in it
        - The CUSTOMER relation has a degree of 3 since it had 3 attributes in it. (cust_id, cust_name, cust_address)
    - Cardinality
        - The cardinality of a relation is the number of tuples (rows) in it
        - If the CUSTOMER relation contains 200 customers, its cardinality is 200
6. Primary key and foreign key
    - Primary Key
        - A primary key is an attribute (or set of attributes) that uniquely identifies each tuples (columns) in a relation
        - Usually shown as the underline in the text representation
        - The primary key in the STUDENT relation is cust_id as it uniquely identifies each customer
    - Foreign Key
        - A foreign key is an attribute in a relation that creates a link between a relation and another relation
        - In the ORDER relation, cust_id is a foreign key that references the order_id in the CUSTOMER relation, this makes a relationship between ORDER and customer relations, which makes it possible to identify which customer placed an order.

1. In any relation, tuples must be unique. However, in many cases, the set of all the attributes in a relation is not considered a candidate key. Why not? On the other hand, suppose we do have a relation where the set of all attributes is a candidate key. In this case, show that this set must, therefore, be the only candidate key and hence the primary key.

A candidate key is a minimal set of attributes (columns) that can uniquely identify a tuple (row) in a table, if all attributes in a relation is considered a candidate key, it does not fulfill the requirements the minimal aspect for it to be a candidate key.

Suppose a relation has a set of all attributes (A, B, C, D) as a candidate key, since there is no smaller subset (A, B, C) which can uniquely identify tuples (rows). This means that all the attributes together make the only candidate key. Since there is only one candidate key, the four attributes that make up the candidate key are the primary key

2. Identify the primary key and foreign key for these three relations:

ORDER (order_id, order_date, cust_id)
ORDERLINE (order_id, prod_no, ol_qtyordered, ol_lineprice)
PRODUCT (prod_no, prod_desc, prod_unitprice)

ORDER
Primary key: order_id
Foreign key: cust_id
ORDERLINE
Primary key: composite key (order_id, prod_no)
Foreign key: order_id, prod_no
PRODUCTLINE
Primary key: prod_no
Foreign key: no foreign key

3. Consider a relation that depicts a dental surgery appointment system.

APPOINTMENT (dentist_id, dentist_name, patient_id, patient_name, appointment_datetime, surgery_roomno)

Identify the superkey(s), candidate key(s) and the primary key for the relation if the following business rules are applicable:

(notes) Superkeys are one or more attributes that can uniquely identify a tuple (row) in a table, it can contain extra attributes (which makes it not a candidate key) that are not necessary for uniqueness.

- A dentist can only see a single patient at a particular date and time
- A dentist treats a patient in a particular surgery room (only one patient is treated in the room at a time), and
- A patient can see the same dentist multiple times.

Superkey
dentist_id, patient_id, appointment_datetime ,surgery_roomno

Candidate key
patient_id, appointment_datetime
or
dentist_id, appointment_datetime

Primary key
dentist_id, appointment_datetime